EV DC Fast Charger – Complete Internal Components Explained

A DC Fast Charging Station is a power-electronics system that converts AC electricity from the grid into controlled DC power and supplies it directly to an EV battery. It also communicates with the vehicle, monitors safety, controls charging current and voltage, and protects both the vehicle and the charger.

This explanation is in English, with component names, functions, connections, and practical examples to help you prepare for an EV / Automotive Embedded Systems internship at AutoMech.

1. DC Fast Charger – Internal Architecture



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The following is a simplified electrical block diagram of a typical DC fast charger.

AC Grid Supply

3-phase AC · typically 400–415 V in India

Input Protection & Switching

MCCB / fuses · surge protection · contactor

EMI Filter + AC/DC Rectifier

Filters electrical noise and converts AC to DC

Power Factor Correction (PFC)

Improves input power factor and shapes grid current

DC Link / DC Bus

Capacitors · voltage sensing · pre-charge circuit

Isolated DC/DC Converter

Regulates DC output and provides galvanic isolation where designed

Output Filter + DC Contactors

Current / voltage sensing · insulation monitoring

EV Battery

Receives controlled DC power through the charging inlet

The control system operates alongside the entire power path:

- Main controller: coordinates charging and safety.
- Communication interface: exchanges charging requirements with the EV.
- Sensors: measure voltage, current, temperature and other conditions.
- Auxiliary power supply: powers controllers, relays, fans and communication circuits.
- Thermal management: keeps power modules and cables within safe temperature limits.

Actual architectures vary. Some chargers combine the rectifier and PFC into one power stage, while others use modular power units in parallel.

2. Complete internal component list

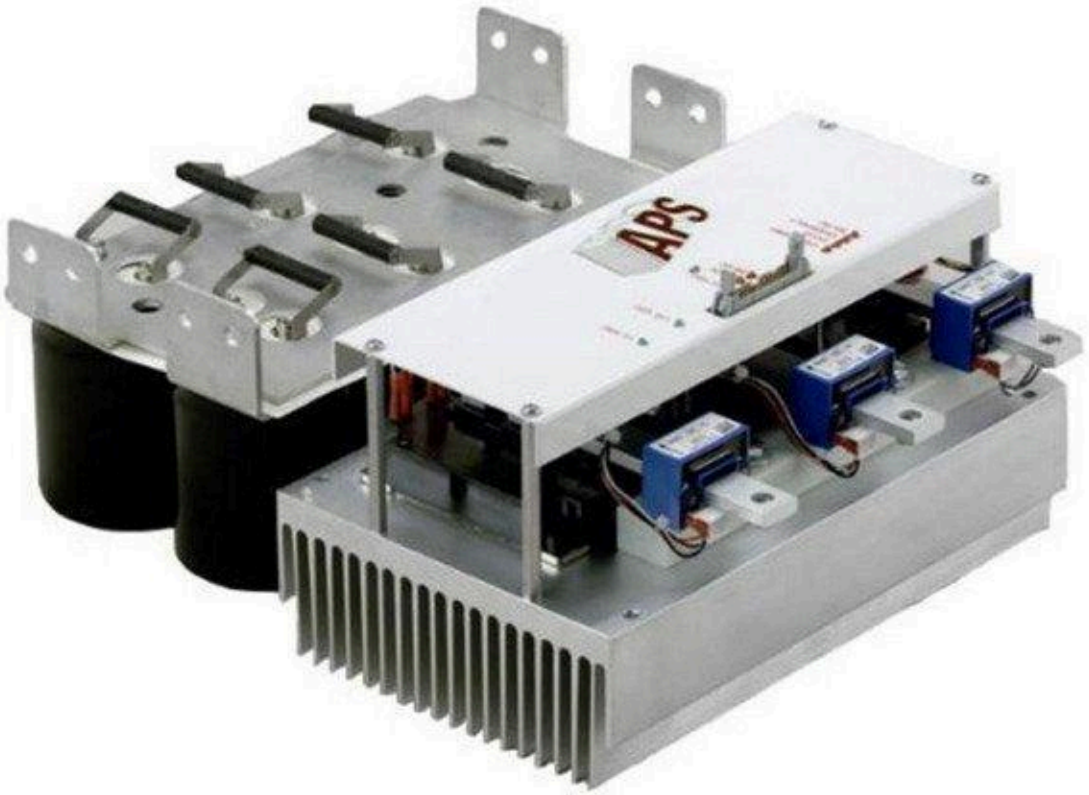
The components below are typical of a commercial DC fast charger. Exact parts, ratings and circuit topology depend on the manufacturer and power rating.



1. Input protection and switching

- MCCB / MCB: protects against overloads and short circuits, according to its rating and application.
- Fuse: interrupts excessive fault current.
- SPD (Surge Protection Device): limits transient voltage surges.
- AC contactor: electrically operated switch for connecting or disconnecting a circuit.
- EMI/RFI filter: reduces conducted electrical noise.

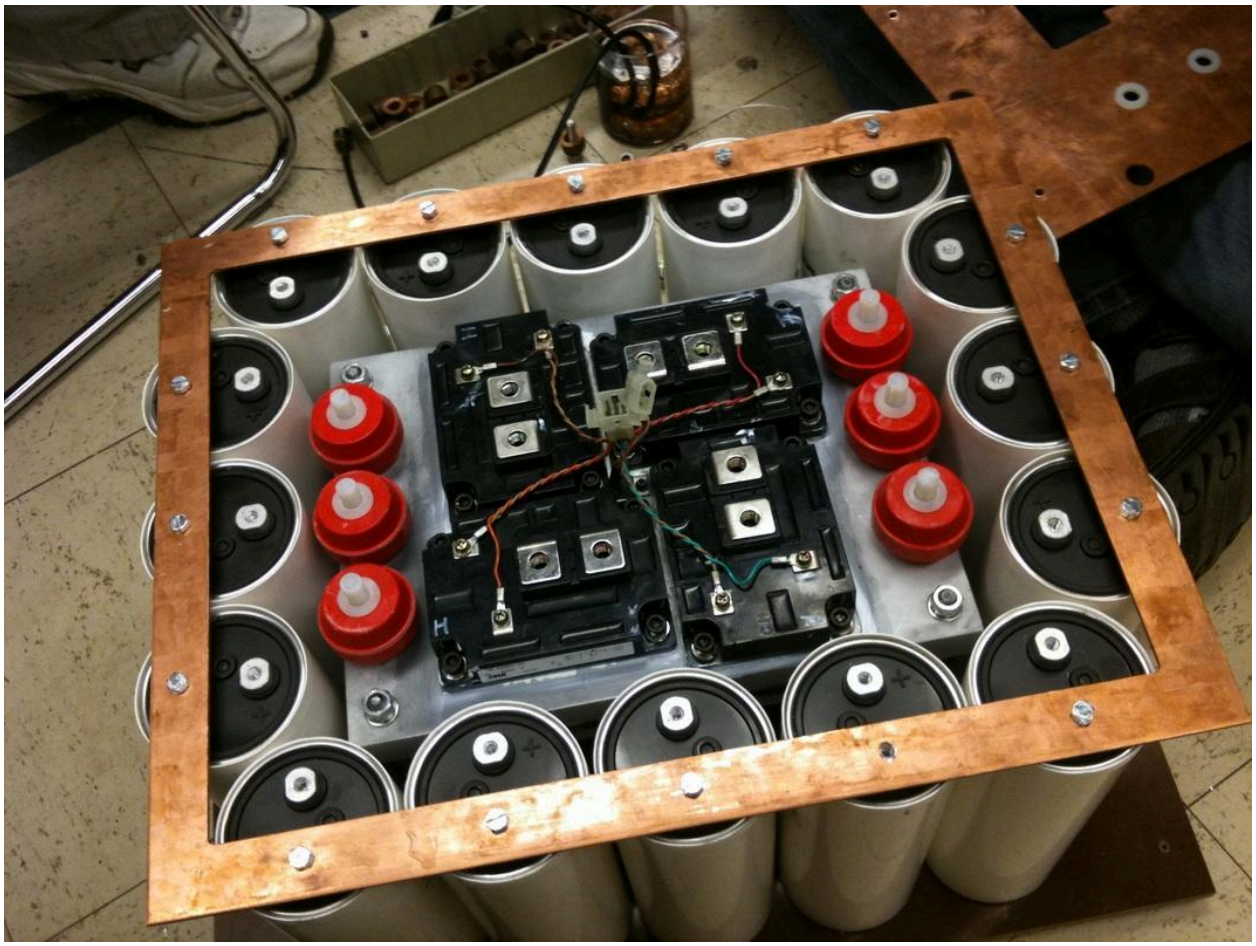
Purpose: Protect the incoming electrical system and charger.



2. AC-to-DC rectifier

- Diode bridge or active semiconductor rectifier.
- IGBT or SiC MOSFET switching devices in active designs.
- Gate-driver circuit.
- Heat sink and temperature sensors.

Purpose: Converts three-phase AC into DC. An active rectifier can also help control the input current and power factor.



3. DC-link circuit

- High-voltage electrolytic or film capacitors.
- DC busbars.
- Voltage measurement circuit.
- Bleeder or discharge resistors.
- Pre-charge resistor and switching circuit, where required.

Purpose: Smooths DC voltage, stores a small amount of electrical energy and stabilizes the intermediate DC bus. It is not the EV's main battery.



4. DC/DC converter

- High-frequency transformer in an isolated topology.
- MOSFETs, IGBTs or SiC MOSFETs.
- Gate drivers.
- Transformer and inductors.
- Rectification stage and output capacitors.

Purpose: Converts the intermediate DC voltage into a controllable output voltage and current. An isolated topology provides galvanic isolation between input and output.



5. Output switching and protection

- High-voltage DC contactors.
- DC-rated fuses.
- Current and voltage sensors.
- Insulation monitoring device (IMD), where required.
- Discharge and fault-detection circuits.

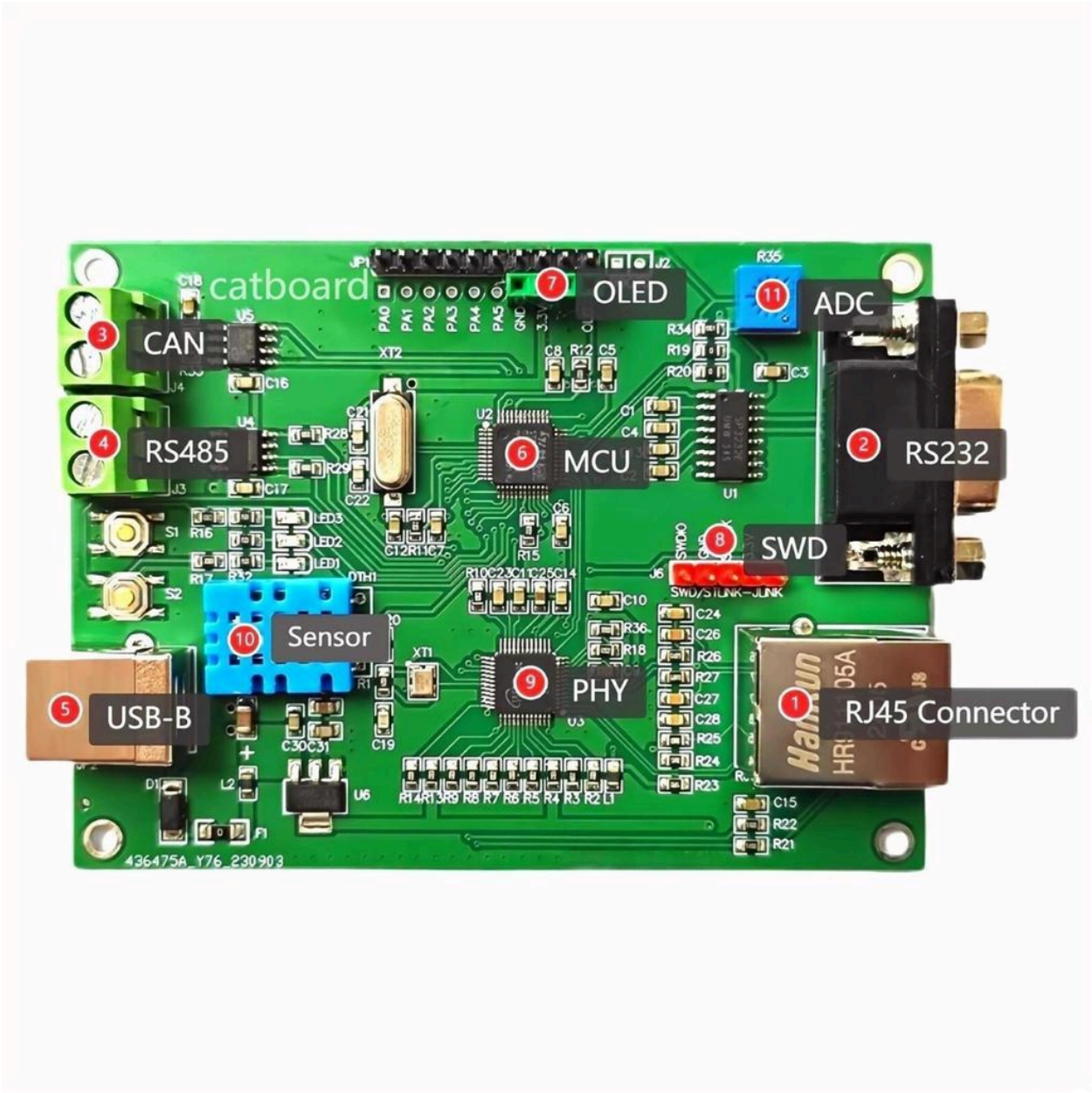
Purpose: Connects the power output safely to the vehicle, monitors faults and disconnects power when required.



6. Charging connector and cable

- CCS2 connector in many Indian passenger-EV applications.
- DC positive and negative contacts.
- Protective earth and communication contacts.
- Temperature sensors in applicable designs.
- Cable, connector locking and release mechanisms.

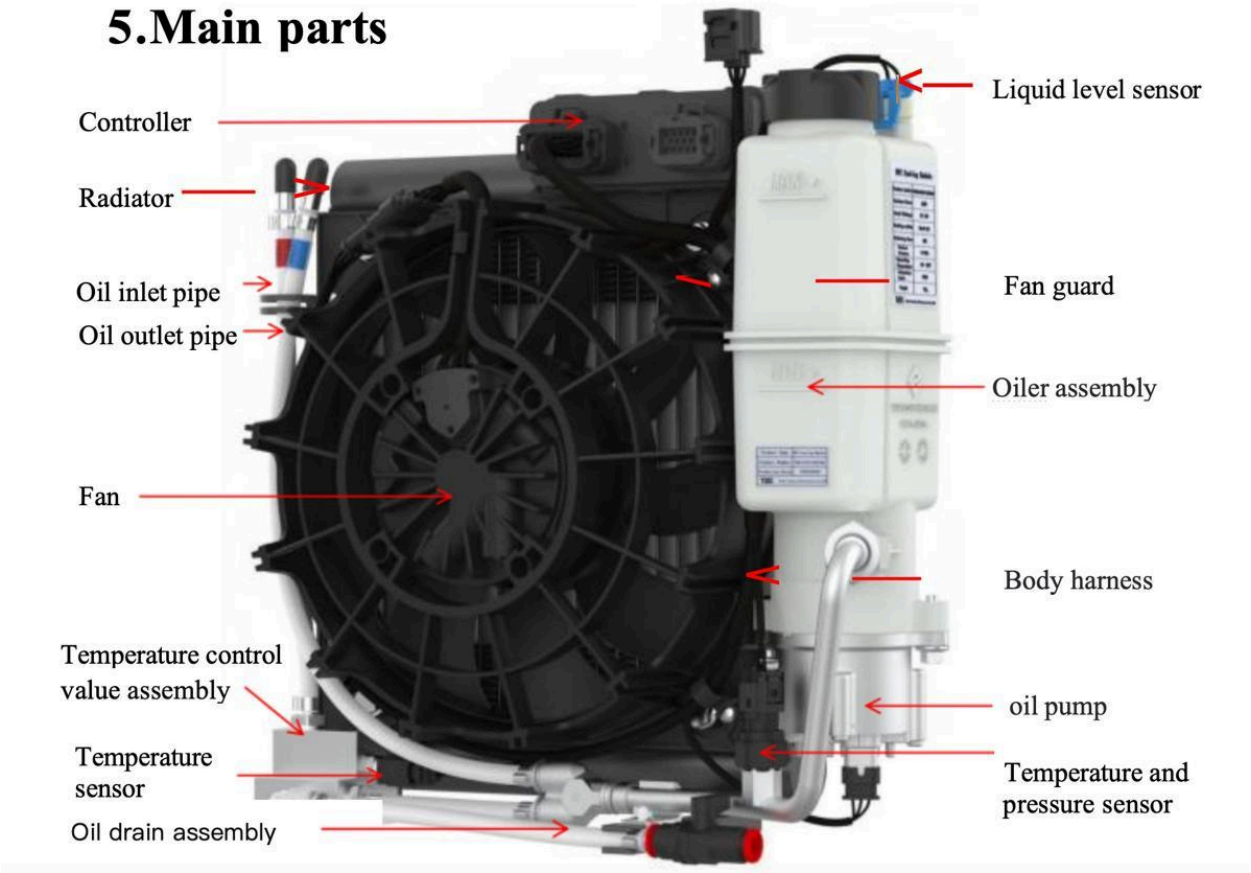
Purpose: Delivers DC power and enables vehicle identification, communication and interlocking. Connector type depends on the vehicle and charging standard.



7. Main controller and communication boards

- MCU or microprocessor.
- ADC and digital input/output circuits.
- CAN, RS-485 or Ethernet interfaces as appropriate.
- PLC communication modem for CCS high-level communication.
- Isolated gate-control and sensor interfaces.
- Watchdog and fault-logging memory.

Purpose: Coordinates the power modules, safety checks, charging sequence and vehicle communication.

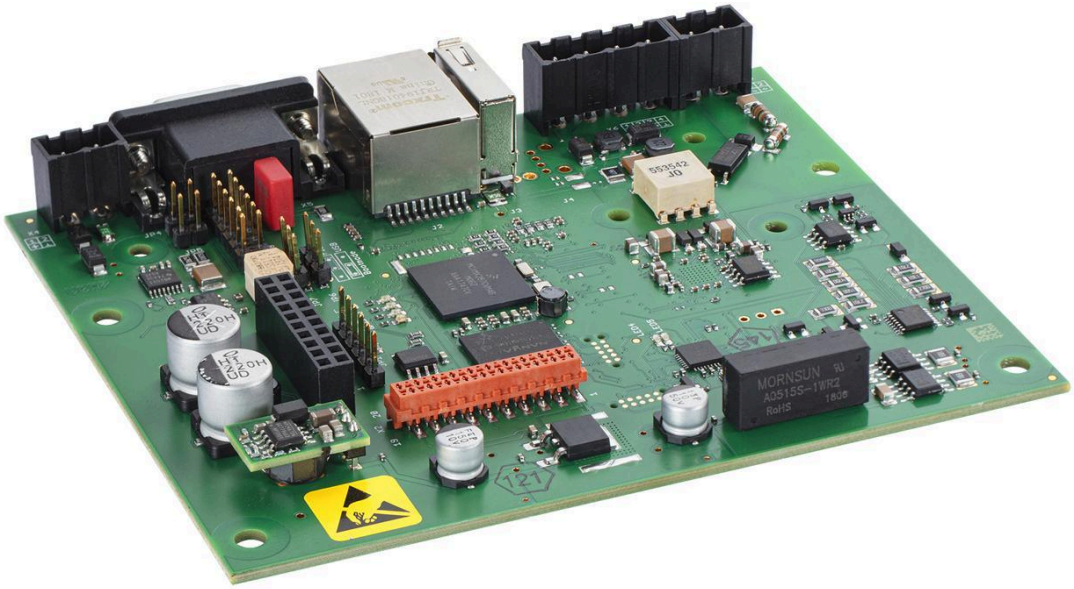


8. Cooling and auxiliary power

- Fans and heat sinks.
- Coolant pump, radiator or heat exchanger in liquid-cooled designs.
- Auxiliary AC/DC power supply.
- Low-voltage DC rails, such as 12 V or 24 V, depending on design.
- Temperature, flow and leak sensors where applicable.

Purpose: Keeps semiconductor devices, cables and control electronics within their specified operating limits.

3. Controller board – what is actually on the PCB?



chargebyte.com

For an embedded hardware internship, you should understand these components particularly well.

PCB component	Function
MCU / processor	Executes the charging-control firmware
ADC	Reads analog sensor voltages
Voltage sensor interface	Measures DC bus and output voltage
Current sensor interface	Measures charging current

CAN transceiver	Provides a physical interface for CAN communication, when used
RS-485 transceiver	Provides differential serial communication, when used
Ethernet PHY	Interfaces the processor with Ethernet
PLC modem	Enables CCS high-level communication over the charging cable
Isolated DC/DC supply	Provides isolated low-voltage power rails
Optocoupler / digital isolator	Transfers signals across an isolation barrier
Gate-driver interface	Controls power semiconductor switching
Relay drivers	Operate relays and contactors
Watchdog	Detects certain firmware execution failures and helps initiate recovery
EEPROM / Flash	Stores configuration, firmware or event information
Temperature-sensor interface	Monitors power modules, connectors or coolant

Important: CAN, RS-485 and PLC are not interchangeable. In a typical CCS charging system, high-level EV-to-charger communication uses PLC technology and standards such as ISO 15118 or DIN 70121. CAN may be used inside the charger or vehicle for other purposes, but it is not a substitute for the CCS PLC link.



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4. How the power electronics work

A. AC-to-DC rectification

The Indian three-phase supply is commonly around 400–415 V AC line-to-line, subject to the installation.

A three-phase rectifier converts AC into DC. For an ideal six-pulse diode bridge, the average rectified voltage is approximately:

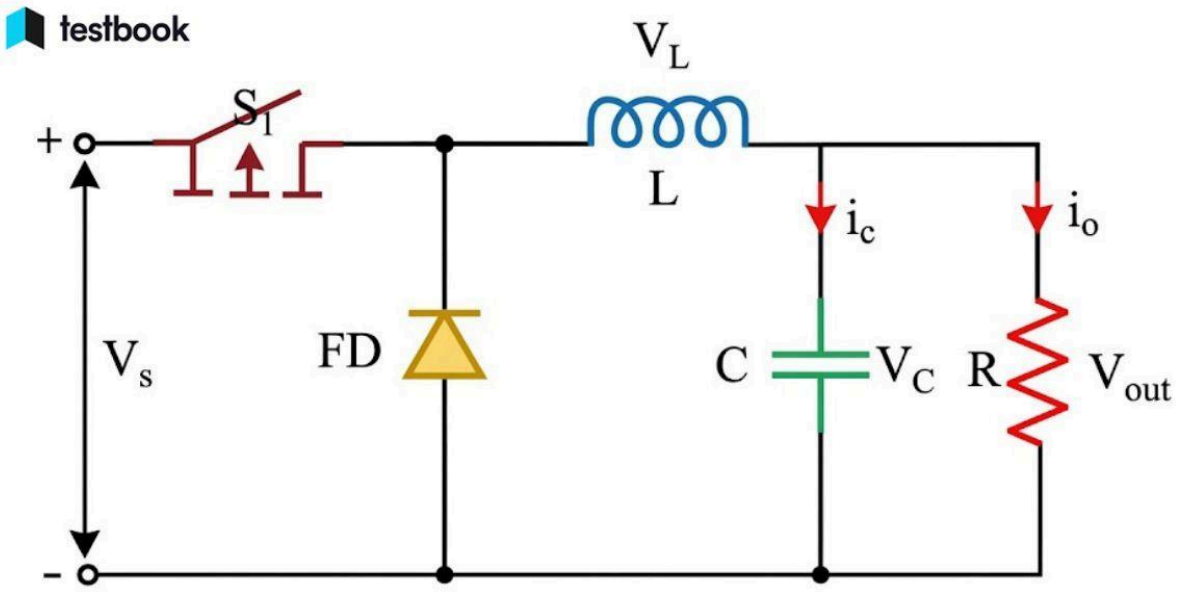
$$V_{DC} \approx 1.35 V_{LL}$$

For a 400 V AC input:

$$V_{DC} \approx 1.35 \times 400 = 540 V$$

This is an illustrative idealized value, not a guaranteed charger DC-link voltage. Real voltage depends on topology, loading, power-factor correction and switching control.

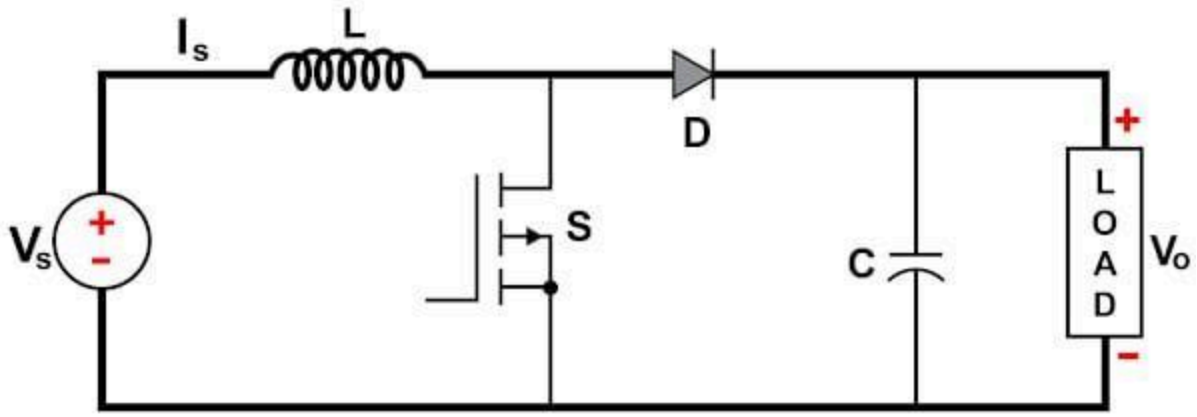
B. DC/DC conversion: buck versus boost



Buck Converter Circuit Diagram

Buck converter

Reduces voltage. For example, a higher DC voltage can be stepped down to a lower DC voltage.



Boost converter

Raises voltage. It is used when the required output voltage is higher than the input voltage.

A commercial DC fast charger may use more advanced topologies, such as an isolated dual-active-bridge or LLC converter, rather than a simple buck or boost circuit.

The converter must regulate its output according to the battery's requested voltage and current, while respecting the charger's own limits.

C. How charging power is calculated

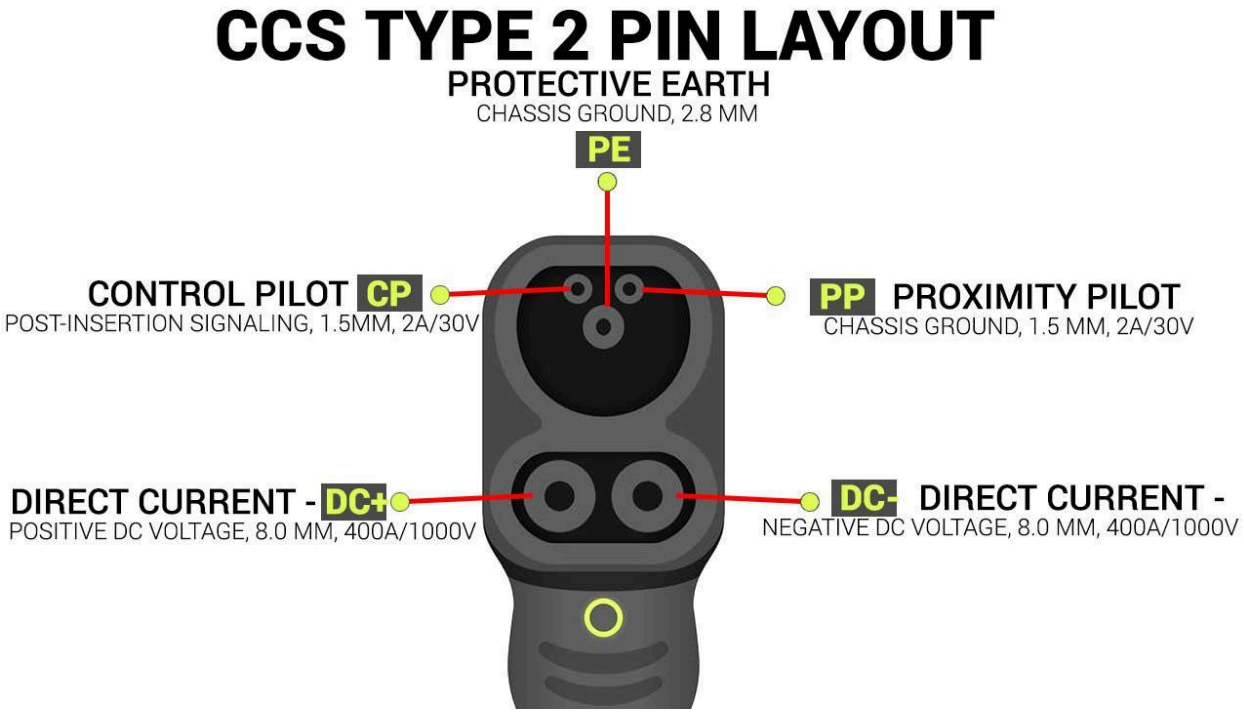
$$P = V \times I$$

Example	Voltage	Current	Power
Lower-power DC charging	400 V	50 A	20 kW

Medium-power charging	500 V	100 A	50 kW
High-power charging	800 V	200 A	160 kW

These are arithmetic examples, not guaranteed operating points. The actual charging power is limited by the charger, vehicle, battery temperature, state of charge, cable and connector ratings.

5. How the charger communicates with the EV



interchargers.com

For CCS charging, two systems work together:

- Physical and basic signalling: connector detection, proximity information, control-pilot signalling and related safety functions.
- High-level digital communication: charging parameters and control messages using PLC-based communication, with standards such as ISO 15118 and DIN 70121.

The exact message sequence depends on the supported protocol version.



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EV / Battery Management System (BMS)

Reports charging requirements and battery limits through the vehicle's charging communication system.

CCS communication interface

Control pilot + PLC / high-level protocol

EVSE / Charger controller

Checks limits, coordinates power modules and commands the charging process.

Rectifier + DC/DC power modules

Regulated DC output to the battery

Typical charging sequence

1. Plug in: The connector is inserted and the charger detects its connection and locking state.
2. Safety checks: The system verifies relevant interlocks and checks that the charging path is safe to energize.
3. Communication setup: The vehicle and charger establish the supported charging communication session.
4. Parameter exchange: The vehicle provides relevant charging limits and requests; the charger communicates its available capabilities.
5. Power delivery: The charger enables its power stages and regulates output voltage and current according to the charging session.
6. Continuous monitoring: The controller monitors current, voltage, temperatures, communication status and faults.
7. Stop charging: When charging is complete or a stop/fault condition occurs, the charger reduces or stops power and safely terminates the session.

For further technical study, CharIN provides CCS technical guides and implementation information: CharIN CCS specification and technical resources.



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6. Safety systems and fault protection

A DC fast charger can operate at hundreds of volts and deliver hundreds of amperes. Its protection system is as important as its power electronics.

Protection component	What it does
MCCB / fuse	Protects against excessive current and short circuits
Ground-fault protection	Detects applicable earth-leakage or ground-fault conditions
Insulation monitoring device (IMD)	Monitors insulation resistance in applicable DC systems
DC contactor	Connects or disconnects the DC power path
Pre-charge circuit	Limits initial charging current into capacitors where required
Voltage sensor	Detects abnormal voltage
Current sensor	Detects overcurrent and measures charging current

Temperature sensor	Detects overheating
Emergency stop	Requests an emergency shutdown
Interlock circuit	Detects specified unsafe access or connector conditions
Watchdog	Helps detect certain controller failures
Coolant flow / leak monitoring	Protects applicable liquid-cooled systems

Example fault: If the output current exceeds the permitted limit, the control system can command the power modules to reduce or stop output and open the appropriate contactors. The precise response depends on the fault, hardware design and safety requirements.

Do not open or probe a live high-voltage charger without the appropriate training, isolation, discharge verification and safety procedures.

7. A practical example: 60 kW DC fast charger



wattsfuture.com

Imagine a charger rated at 60 kW with a DC output range of 200–1000 V. This is an illustrative specification, not a claim about a particular commercial model.

At 400 V output, the current needed to deliver 60 kW is:

$$I = \frac{P}{V} = \frac{60{,}000}{400} = 150A$$

At 800 V output, the same power requires:

$$I = \frac{60{,}000}{800} = 75A$$

In practice, the charger must stay within its rated voltage, current, cable and thermal limits. The EV battery may request less than the full 60 kW.

8. Which components should you learn for an embedded hardware internship?

For a role involving charger controllers, BMS, VCU, schematics and PCB testing, focus on these in order:

Priority 1 — Controller PCB

MCU, ADC, GPIO, isolated power supplies, CAN transceivers, RS-485, Ethernet and sensor interfaces.

Priority 2 — Measurement and debugging

Multimeter, oscilloscope, voltage dividers, current sensors, continuity testing, power-rail checks and fault diagnosis.

Priority 3 — Power electronics

Rectifiers, MOSFETs, IGBTs, gate drivers, inductors, transformers, buck/boost converters and DC-link capacitors.

Priority 4 — Charging communication

CCS2 connector, control pilot, PLC, ISO 15118, DIN 70121 and charging-session states.

Priority 5 — Schematics and PCB design

Learn to read reference schematics, identify component values and ratings, trace signals, prepare a BOM, and perform basic PCB design and bring-up.

9. Free technical resources

- CharIN — CCS technical specifications — charging communication and implementation.
- CharIN — CCS basic technical details — technical guides and implementation documents.
- Texas Instruments — search for EV charging, isolated gate drivers, current sensing and DC/DC converters.
- STMicroelectronics — search for power conversion, STM32 motor control and isolated power supplies.
- KiCad — free schematic capture and PCB design software.

10. Check your understanding

1. Which stage converts the grid AC supply into DC?

DC contactor
Rectifier
Current sensor

2. Which technology normally carries CCS high-level communication between the EV and charger?

RS-485 only
Bluetooth
PLC

3. What is the primary purpose of a DC/DC converter?

Regulate DC voltage/current
Display charging prices
Lock the connector

4. Which component monitors insulation resistance in applicable DC charging systems?

IMD
Ethernet PHY
MCCB alone
Reset quiz

The best next step is to study a DC fast charger controller reference schematic, component by component, and then trace how the controller commands the power stage and communicates with the EV. That connects power electronics, embedded firmware and PCB debugging into one practical learning path.